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Operation FlutterBridge: macOS Malvertising Campaign Spreads New FlutterShell Backdoor

Malware Analysis

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TLP:CLEAR | Lost Boys Cyber | Operation FlutterBridge: macOS Malvertising Campaign Spreads New FlutterShell Backdoor - Malware Analysis

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1. Executive Summary

Operation FlutterBridge is a macOS malvertising campaign attributed by Unit 42 to activity cluster CL-CRI-1089. The campaign uses hundreds of Google-verified ads and shell-company identities to lure users into downloading notarized desktop applications that appear to be benign podcast or PDF utilities. Once installed, those applications deliver FlutterShell, a backdoor built with the Flutter framework that keeps its core malicious logic off-binary and retrieves attacker-controlled JavaScript at runtime through a WebView bridge.

The campaign is operationally significant for three reasons. First, the malware currently behaves like adware on many infected hosts by hijacking Google Chrome search and new-tab flows to an attacker-controlled domain, but the underlying command set already supports broader backdoor use including arbitrary shell execution, filesystem interaction, and environment-variable exfiltration. Second, the use of valid Apple Developer IDs and successful notarization reduced pre-execution friction and suppressed common user trust signals. Third, the architecture allows the actor to change functionality without redistributing the app, enabling rapid tradecraft shifts while preserving the same signed lure.

Observed variants masqueraded as PodcastsLounge, PDF-Brain, and PDF-Ninja. Unit 42 also documented unfinished functions and inactive code paths, indicating the malware family remained under active development during the reporting period. The latest variants increased string and symbol obfuscation, renamed commands to resemble legitimate PDF-library routines, and exposed an AI summarization workflow that can proxy document contents through attacker infrastructure before forwarding them to a downstream AI service.

Defenders should treat FlutterShell as more than a nuisance browser hijacker. Its present monetization path appears adware-oriented, but its architecture, exfiltration channel, and live command bridge justify triaging it as a backdoor-capable intrusion. Immediate priorities are blocking known infrastructure, auditing macOS application deployments that impersonate media or document utilities, hunting for suspicious modifications to Chrome Secure Preferences, and reviewing outbound traffic for ``/update-thanks.html``, ``/getConfig``, ``/getUpdateThanksConfig``, and ``/summarize-text`` style activity.

2. Sample Metadata

Field	Value
Malware family	FlutterShell
Campaign / cluster	Operation FlutterBridge / CL-CRI-1089
Type	macOS backdoor with adware/browser-hijacking behavior and dynamic JavaScript-controlled command bridge
Primary platform	macOS
Delivery model	Google Ads malvertising leading to signed, notarized fake desktop applications
Masquerades observed	PodcastsLounge, PDF-Brain, PDF-Ninja
Public reporting date	2026-06-02
Primary technical source	Palo Alto Networks Unit 42
Representative sample hash	<code>`30448686ec900d5213d74f08f0d2b7924c5336a29445b2a434aba8d8b19d7530`</code> ('PDF-Ninja.app')
Representative adware domain	<code>`sinterfumesco[.]com`</code>
Key C2 / payload paths	<code>`/update-thanks.html`</code> , <code>`/getConfig`</code> , <code>`/getUpdateThanksConfig`</code> , <code>`/api/update-delay`</code> , <code>`/summarize-text`</code>
Developer IDs observed	Yasar Sever (<code>`UBZDAAV97Y`</code>), Batuhan Dabag

	(`FW9NHQ8922`), Yusuf Bal (`B73CHZ24Y8`)
Targeting emphasis	Anglophone and Western European markets, but globally accessible ad distribution
Confidence	High for architecture, variants, hashes, and domains documented in the Unit 42 source; Medium for victim counts and follow-on objectives beyond observed activity

Operational handling summary:

- Assume compromise can progress from adware-like browser hijacking to broader hands-on-backdoor activity without a new binary appearing on disk.
- Preserve both the application bundle and related website content when triaging because key logic is loaded remotely at runtime.
- Treat notarization and a valid Apple Developer signature as insufficient trust indicators for this campaign.

3. Source Review and Confidence

Source	Contribution	Confidence
Palo Alto Networks Unit 42, "Operation FlutterBridge: macOS Malvertising Campaign Spreads New FlutterShell Backdoor"	Primary source for campaign narrative, infection chain, hashes, domains, bundle IDs, developer IDs, and technical behavior.	High
Moonlock / Moonlock Labs reporting on JSCoreRunner fake PDF-converter malware	Secondary corroboration that the same ecosystem previously used fake PDF utilities and related distribution themes on macOS.	Medium
The Hacker News coverage of FlutterShell	Secondary corroboration that the campaign and backdoor were externally observed and rapidly amplified across the security community.	Low-Medium
MITRE ATT&CK knowledge base	Technique normalization for defender-oriented mapping and detection structuring.	High

Confidence assessment:

- High confidence that FlutterShell uses a WebView-based JavaScript-to-native bridge, delayed callback behavior, signed/notarized lure apps, and Chrome Secure Preferences modification because these behaviors are described with concrete implementation details in the primary source.
- High confidence in the three named macOS lure variants, their bundle IDs, and their listed SHA256 values because the source published those artifacts directly.
- Medium confidence that the actor's long-term intent remains browser hijacking alone. The command set and document-proxy summarization design support a broader backdoor and data-exfiltration role than the currently dominant adware behavior.
- Medium confidence in campaign scale and exact victim distribution. Unit 42 described hundreds of active advertisements and clear geographic focus areas, but public reporting cannot fully quantify exposure.
- Medium confidence in the exact operational overlap with earlier Windows and JSCoreRunner activity. The infrastructure and behavioral links are compelling, but this report treats them as cluster-level context rather than a claim of identical tooling.

Assessment boundaries:

- This report is source-backed analysis based on publicly available research and locally preserved article evidence. No independent detonation of the macOS samples was required for the intake-review deliverable.
- Defensive guidance therefore emphasizes observable behaviors and published indicators rather than unsupported reverse-engineering claims beyond the cited source.

4.1 Delivery and social-engineering path

Operation FlutterBridge begins with Google Ads that impersonate legitimate end-user utilities. The ads route victims to polished download pages for applications such as PodcastsLounge, PDF-Brain, and PDF-Ninja. The apparent legitimacy is reinforced by functional application behavior, valid Apple Developer signing, and Apple notarization. In practical terms, the actor front-loads trust signals at the exact point where many users would otherwise hesitate.

Variant / Artefact	Delivery Role	Notes
`PodcastsLounge.dmg` / `podcasts_lounge.app`	Earliest observed macOS lure	Podcast-player theme; associated with `com.app.podcastsLounge`
`PDF-Brain.dmg` / `PDF-Brain.app`	Follow-on lure	PDF utility theme; internal naming still exposed links back to `podcasts_lounge`
`PDF-Ninja.dmg` / `PDF-Ninja.app`	Most obfuscated observed variant	Uses Flutter `--obfuscate` and deceptive command names intended to resemble legitimate PDF-library operations

4.2 Runtime execution flow

A representative FlutterShell execution chain is shown below:

1. The victim clicks a malvertising result and downloads a signed, notarized macOS application that appears to be benign.
2. The application launches normally and may provide expected surface functionality, reducing immediate suspicion.
3. FlutterShell contacts an attacker-controlled endpoint such as `/api/update-delay` to obtain a runtime delay before presenting malicious web content.
4. After the delay expires, or when the victim triggers certain UI actions such as About or Update, the app loads attacker-controlled web content in an embedded WebView.
5. The web page injects JavaScript through the `flutterInvoke` bridge, passing JSON-formatted commands from the WebView into the native Dart environment.
6. The native layer translates those commands into local system operations including shell execution, filesystem interaction, environment-variable capture, or browser configuration changes.
7. In the observed monetization path, FlutterShell fingerprints the device, alters Chrome Secure Preferences, terminates Chrome, and relaunches it against the attacker-controlled adware domain `sinterfumesco[.]com`.
8. In newer variants, the same remote-command model also supports AI summarization workflows that relay user document content through attacker infrastructure before delivering summaries back to the user.

4.3 Why the chain matters

- The logic that matters most is delivered at runtime, so a static allow decision on the app bundle alone is not enough.
- Time-delayed execution undermines fast sandbox verdicts and builds user trust before overtly malicious behavior begins.
- The same signed application can change behavior as the operator updates JavaScript or server-side configuration.

5.1 WebView bridge architecture

FlutterShell is built with the Flutter framework and uses a WebView-based architecture rather than embedding most malicious behavior directly in the compiled application. The malware loads attacker-controlled web content and relies on a JavaScript-to-native bridge named `flutterInvoke` to pass JSON commands into native Dart handlers. This approach separates lure presentation from core malicious logic and lets the actor update functionality without shipping a new binary.

This architecture materially changes defender assumptions. Static scanning of the application may reveal only scaffolding, while the runtime website supplies the operative behavior. The source also notes that FlutterShell waits for a server-provided delay before loading attacker content, which supports both sandbox evasion and user-trust shaping.

5.2 Variant progression and obfuscation

Unit 42 documented three primary variants spaced roughly one month apart through early February 2026:

Variant	Primary theme	Notable technical traits	Defensive implication
PodcastsLounge	Podcast player	Earliest named variant; descriptive internal command naming	Best baseline for family linkage and initial hunting
PDF-Brain	PDF viewer	Partial string obfuscation; package naming still linked to podcasts_lounge lineage	Shows iterative hardening without fully severing family artifacts
PDF-Ninja	PDF viewer	Flutter `--obfuscate`, randomized symbols, and deceptive command names	Raises reverse-engineering cost and weakens simple string-based signatures

The actor also reused site design elements across lure families and maintained Windows versions of the same applications, although Unit 42 did not observe equivalent malicious logic in the Windows builds during the reporting window. That pattern is consistent with phased deployment and testing against ecosystem trust controls.

5.3 Backdoor and adware behavior

The primary observed backdoor capabilities published by Unit 42 were:

- arbitrary command execution;
- filesystem interaction;
- environment-variable exfiltration.

Operationally, the campaign was most often seen using those capabilities to support browser hijacking. FlutterShell fingerprints the host by retrieving `IOPlatformUUID`, then modifies the Chrome `Secure Preferences` file to replace `default\_search\_provider\_data` values such as `url` and `new\_tab\_url` with `sinterfumesco[.]com`. It then forcibly terminates and relaunches Google Chrome with command-line arguments that reopen a session and suppress crash-recovery prompts.

This sequence matters because it shows a clear revenue-generating use case while also proving the malware can cross from its Flutter UI into native OS execution. Even if a defender initially classifies the event as adware, the command bridge means the same foothold can later support richer data collection or shell-driven post-compromise actions.

5.4 Document-proxy exfiltration and update abuse

One of the more concerning later-stage features is an AI summarization workflow that proxies document contents through attacker infrastructure. Instead of sending the file directly to an AI assistant endpoint, FlutterShell forwards the contents to the attacker's `/summarize-text` path first, allowing the actor to capture the full document before returning a summary to the victim. This is a materially different risk than adware monetization because it turns normal user productivity activity into covert content exfiltration.

Unit 42 also described abuse of the Sparkle update framework. Rather than waiting for the user to approve a new version, FlutterShell monitors update-completion callbacks, verifies the staged installation directory, programmatically opens the downloaded app bundle, and terminates the old process. That enables silent capability upgrades without user interaction.

5.5 Infrastructure and campaign context

The malware infrastructure observed in the source aligns with an organized malvertising operation rather than a one-off rogue application.

Infrastructure element	Example(s)	Role
------------------------	------------	------

WebView C2 / content hosts	`atsheisdomeestic[.]org`, `etoftheappyrince[.]org`, `healightejustb[.]org`	Hosts delayed content, update pages, and JavaScript command logic
Adware redirector	`sinterfumesco[.]com`	Receives hijacked Chrome searches and new-tab traffic
Advertiser shell entities	AdsParkPro LTD, Advantage Web Marketing LLC, SOFT WE ART LIMITED	Enable ad-buying, publisher verification, and operational continuity
Related cluster designation	CL-CRI-1089	Links FlutterShell to broader malvertising activity and prior macOS/Windows campaigns

From a defender perspective, the most durable campaign signal is the combination of signed-but-fraudulent app lures, delayed WebView retrieval, bridge-based command execution, Chrome preference tampering, and shell-company-backed ad distribution.

6. Indicators of Compromise

The complete indicator set is preserved in `iocs.csv`. The subset below is the most operationally useful for rapid triage and blocking.

Type	Value	Description
sha256	`021666417de8b9972c179783fe60d4c4ad2d93224e3a0f16137065c960b1b845`	`PodcastsLounge.dmg`
sha256	`363923500ce942bf1a953e8a4e943fbf1fb1b5ed6e5d247964c345b3ad5bfc34`	`podcasts_lounge.app`
sha256	`9053e8ddaecca1f960c041c944ca8799fc71dc86a4b50d2639ee4e0d2cb82f47`	`PDF-Brain.app`
sha256	`30448686ec900d5213d74f08f0d2b7924c5336a29445b2a434aba8d8b19d7530`	`PDF-Ninja.app`
sha256	`48047c34bbd57fe1e24bc538bc2ce9e0ac4c4eb48d3b0c195b414f0379dc0745`	`PDF-Ninja` associated dynamic library
domain	`atsheisdomeestic[.]org`	PodcastsLounge C2 / update-thanks host
domain	`etoftheappyrince[.]org`	PDF-Brain C2 / update-delay host
domain	`healightejustb[.]org`	PDF-Ninja C2 / JavaScript payload host
domain	`sinterfumesco[.]com`	Browser-hijack / adware redirect domain
path	`/update-thanks.html`	WebView-delivered payload path
path	`/getConfig`	Remote configuration / JavaScript logic endpoint
path	`/getUpdateThanksConfig`	Alternate payload / configuration endpoint
path	`/api/update-delay`	Delay-calculation endpoint

path	`/summarize-text`	AI-summary relay / document-exfiltration path
bundle_id	`com.app.podcastsLounge`	PodcastsLounge app bundle
bundle_id	`com.app.pdfBrain`	PDF-Brain app bundle
bundle_id	`com.pdfninja.app`	PDF-Ninja app bundle
developer_id	`UBZDAAV97Y`	Apple Developer ID for Yasar Sever
developer_id	`FW9NHQ8922`	Apple Developer ID for Batuhan Dabag
developer_id	`B73CHZ24Y8`	Apple Developer ID for Yusuf Bal

Additional triage pivots:

- Commands referencing `IOPlatformUUID`, `killall "Google Chrome"`, or Chrome relaunch flags such as `--restore-last-session` and `--hide-crash-restore-bubble`.
- WebView bridge references to `flutterInvoke`.
- Sparkle update paths under `\$HOME/Library/Caches/com.app.[appname]/org.sparkle-project.Sparkle/Installation/`.

7. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1189	Drive-by Compromise	Victims are drawn from web advertisements into malicious download sites and browser-delivered runtime content.
T1036	Masquerading	The malware poses as legitimate podcast and PDF utilities and uses deceptive naming to blend with expected software behavior.
T1059.007	Command and Scripting Interpreter: JavaScript	Attacker-controlled web content drives local behavior through a JavaScript bridge in the embedded WebView.
T1059.004	Command and Scripting Interpreter: Unix Shell	FlutterShell supports arbitrary shell command execution and uses native command invocation on macOS.
T1082	System Information Discovery	The malware fingerprints systems using `IOPlatformUUID` and collects environment variables.
T1083	File and Directory Discovery	Published command capabilities include filesystem interaction and directory access.
T1105	Ingress Tool Transfer	Runtime logic and update content are fetched from attacker-controlled endpoints after installation.
T1567	Exfiltration Over Web Service	The `/summarize-text` workflow routes

		victim document contents through attacker infrastructure before any downstream AI processing.
T1497.003	Virtualization/Sandbox Evasion: Time Based Evasion	The malware waits for a server-supplied delay before contacting the malicious web payload.
T1027	Obfuscated/Compressed Files and Information	Later variants use stronger Flutter obfuscation and deceptive command naming to complicate analysis.

Interpretation:

- The ATT&CK pattern is strongest around disguise, web-mediated execution, delayed runtime activation, and opportunistic exfiltration.
- Public reporting did not establish durable persistence beyond the application itself, so this report avoids overstating persistence techniques.

8.1 Detection logic summary

Signal	Telemetry Source	Analytic Goal
Unsigned expectation mismatch: newly installed podcast/PDF utility on macOS followed by external traffic to atypical `.org` domains or `/update-thanks.html` paths	EDR process + network telemetry, proxy, DNS	Surface suspicious utility-themed application launches that immediately fetch remote logic
Chrome Secure Preferences write followed by `killall "Google Chrome"` and forced browser relaunch	EDR file/process telemetry on macOS	Detect the observed browser-hijack workflow
Child shell or native command execution from PodcastsLounge, PDF-Brain, PDF-Ninja, or similarly themed app bundles	EDR process telemetry	Identify abuse of FlutterShell's arbitrary command capability
Requests to `/api/update-delay`, `/getConfig`, `/getUpdateThanksConfig`, or `/summarize-text` on recently seen lure domains	Proxy, DNS, network sensors	Detect delayed payload fetches and document-proxy exfiltration
Sparkle cache install path activity followed by silent app relaunch	File telemetry + process telemetry	Catch update-mechanism abuse that upgrades the malware without user approval

8.2 Example Sigma rule

```

title: Suspicious macOS Browser Hijack Commands From FlutterShell-Themed Applications
id: 38d4b2db-2e77-47a7-a274-d5f556b3d4ec
status: experimental
description: Detects macOS utility-themed applications documented in Operation FlutterBridge spawning host-fingerprinting or Chrome-hijack commands.
references:
  - https://unit42.paloaltonetworks.com/flutterbridge-new-fluttershell-backdoor/
author: Lost Boys Cyber
logsource:
  category: process_creation
  product: macos
detection:

```

```

selection_parent:
  ParentImage|contains:
    - 'PodcastsLounge.app'
    - 'PDF-Brain.app'
    - 'PDF-Ninja.app'
selection_cmd:
  CommandLine|contains:
    - 'IOPlatformUUID'
    - 'killall "Google Chrome"'
    - '--restore-last-session'
    - 'Secure Preferences'
condition: selection_parent and selection_cmd
falsepositives:
  - Very rare; investigate any productivity-themed app bundle issuing these commands.
level: high

```

8.3 Example KQL analytic

```

// FlutterShell macOS detection: suspicious app launch + Chrome preference tamper or attacker path
access
let suspiciousApps = dynamic(["PodcastsLounge", "PDF-Brain", "PDF-Ninja"]);
let suspiciousDomains = dynamic(["atsheisdomestic.org", "etoftheappyrince.org", "healightejustb.org",
"sinterfumescos.com"]);
let suspiciousPaths = dynamic(["/update-thanks.html", "/getConfig", "/getUpdateThanksConfig",
"/api/update-delay", "/summarize-text"]);
let proc = DeviceProcessEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where InitiatingProcessFileName has_any (suspiciousApps)
  or FileName in~ ("ioreg", "killall", "open")
  or ProcessCommandLine has_any ("IOPlatformUUID", "Google Chrome", "--restore-last-session", "Secure
Preferences")
| project DeviceId, DeviceName, ProcTime=Timestamp, FileName, ProcessCommandLine,
InitiatingProcessFileName, InitiatingProcessFolderPath;
let filemods = DeviceFileEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where FileName =~ "Secure Preferences"
| project DeviceId, FileTime=Timestamp, FileName, FolderPath, InitiatingProcessFileName;
let net = DeviceNetworkEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where RemoteUrl has_any (suspiciousDomains) or RemoteUrl has_any (suspiciousPaths)
| project DeviceId, NetTime=Timestamp, RemoteUrl, InitiatingProcessFileName,
InitiatingProcessCommandLine;
proc
| join kind=leftouter filemods on DeviceId
| join kind=leftouter net on DeviceId
| where (isnull(FileTime) or FileTime between (ProcTime - 30m .. ProcTime + 2h))
  and (isnull(NetTime) or NetTime between (ProcTime - 30m .. ProcTime + 24h))
| order by ProcTime desc

```

9.1 Hunt hypotheses

- Hosts that installed utility-themed macOS apps from search ads may have reached attacker infrastructure even if no subsequent malware alert fired, because core logic is fetched over the network after launch.
- macOS endpoints that modified Chrome Secure Preferences and then relaunched Chrome unexpectedly may represent successful FlutterShell browser hijacking.
- Users of fake PDF-converter or document-helper applications may have exposed document contents through the actor's `/summarize-text` relay even when no browser-hijack indicators were recorded.

9.2 Hunt query: Chrome tamper + suspicious network correlation

```

let suspiciousDomains = dynamic(["atsheisdomestic.org", "etoftheappyrince.org", "healightejustb.org",
"sinterfumescos.com"]);
let prefs = DeviceFileEvents

```

```

| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where FileName =~ "Secure Preferences"
| project DeviceId, DeviceName, PrefTime=Timestamp, FolderPath, InitiatingProcessFileName;
let proc = DeviceProcessEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where ProcessCommandLine has_any ("Google Chrome", "--restore-last-session", "--hide-crash-restore-
bubble", "IOPlatformUUID")
| project DeviceId, ProcTime=Timestamp, FileName, ProcessCommandLine, InitiatingProcessFileName;
let net = DeviceNetworkEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where RemoteUrl has_any (suspiciousDomains)
| project DeviceId, NetTime=Timestamp, RemoteUrl, InitiatingProcessFileName;
prefs
| join kind=leftouter proc on DeviceId
| join kind=leftouter net on DeviceId
| where (isnull(ProcTime) or ProcTime between (PrefTime - 30m .. PrefTime + 2h))
    and (isnull(NetTime) or NetTime between (PrefTime - 30m .. PrefTime + 24h))
| order by PrefTime desc

```

9.3 Hunt query: suspicious document-summary relay behavior on macOS

```

let lureApps = dynamic(["PodcastsLounge", "PDF-Brain", "PDF-Ninja"]);
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where OSPlatform =~ "macOS"
| where InitiatingProcessFileName has_any (lureApps)
    or InitiatingProcessCommandLine has_any (lureApps)
| where RemoteUrl has_any ("/summarize-text", "/getConfig", "/getUpdateThanksConfig", "/update-
thanks.html")
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, RemoteUrl,
RemoteIP
| order by Timestamp desc

```

Hunt interpretation:

- Prioritize results where file modifications, browser relaunches, and suspicious network destinations cluster on the same host.
- Preserve the full application bundle and supporting web content for any positive result; remote logic retrieval is part of the evidence chain.

10. Recommendations

Priority	Owner	Action
Critical	Endpoint / SOC	Block and alert on the domains and paths in `iocs.csv`, especially `sinterfumesco[.]com`, `/update-thanks.html`, `/getConfig`, `/getUpdateThanksConfig`, `/api/update-delay`, and `/summarize-text`.
High	IT / Endpoint Engineering	Review recent macOS software installations for PDF, podcast, media, or conversion tools that were delivered via ads or unapproved download sites; remove and reimagine affected hosts where FlutterShell artifacts are confirmed.
High	Threat Hunting	Query for Chrome `Secure Preferences` modification followed by Chrome termination/relaunch on macOS. Treat any

		such sequence as suspicious until explained.
High	Identity / IR	For confirmed infections, assume browser session and document-content exposure. Reset browser-backed sessions, review SSO tokens, and assess sensitive documents processed through fake summarization features.
Medium	Security Engineering	Add detections for utility-themed app bundles spawning `ioreg`, `killall`, or `open` with Chrome-related command lines, and baseline Sparkle update behavior for sanctioned internal apps.
Medium	Awareness / IT	Educate users that Apple notarization and a valid signature do not guarantee legitimacy when software is obtained through ads or lookalike download portals.

10.1 Triage checklist

- Isolate hosts that contacted the listed domains or executed the listed app bundles.
- Collect the application bundle, Chrome profile data, relevant EDR telemetry, and outbound HTTP/S metadata.
- Inspect Chrome `Secure Preferences` for unauthorized `default\_search\_provider\_data`, `url`, or `new\_tab\_url` changes.
- Review shell history, child-process telemetry, and Sparkle cache directories for evidence of silent upgrades or follow-on execution.
- If `/summarize-text` activity is present, identify documents processed through the fake application and treat them as potentially exposed.

11. References

9. Palo Alto Networks Unit 42, "Operation FlutterBridge: macOS Malvertising Campaign Spreads New FlutterShell Backdoor," 2026-06-02. <https://unit42.paloaltonetworks.com/flutterbridge-new-fluttershell-backdoor/>
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13. MITRE ATT&CK, Technique T1189: Drive-by Compromise. <https://attack.mitre.org/techniques/T1189/>